Project Proposal

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0.1 Introduction

For this project I am going to focus on a result from elementary number theory as I am accustomed to working with natural numbers and inductive proofs in lean. I will aim to formalise the existence part of the fundamental theorem of arithmetic, i.e. for any natural number $n > 1$, there exists prime numbers $p_1, p_2, \ldots, p_k$ such that $n = p_1 p_2 \cdots p_k$. If I have time, or if I complete all the lemmas and the main result in under 200 lines, I might try to also prove that the primes are unique up to reordering.

0.2 Proof Outline

The basic proof I will formalise will be a proof by induction as follows:

- If $n > 1$ is prime, then its factorisation is itself ($n = 2$ is the base case)
- If $n > 2$ is composite then it has a factorisation $n = ab$ with $1 < a, b < n$
- Apply the induction hypothesis to get $a = p_1 \cdots p_k$ and $b = q_1 \cdots q_l$
- Combine the factorisations of a and b to get a factorisation for n

0.3 Definitions

Here are the basic definitions I will use:

- I will use Mathlib's definition for primality
- I will use my own definition of a prime factorisation of n as simply a list of natural numbers that are primes and also whose product is equal to n

0.4 Results

Here are the lemmas I intend to prove: (more will probably be required)

1. Every prime number has a prime factorisation (a singleton list containing itself)
2. An empty list represents a prime factorisation of one
3. Appending two prime factorisations gives another prime factorisation
4. Every natural number greater than one is either prime or has a factorisation into two natural numbers larger than one and smaller than itself
5. Strong induction for the main result

0.5 Mathlib Usage

As mentioned earlier, I will use the primality predicate from Mathlib as a definition. I will also use Mathlib's definition of natural numbers and lean's builtin linked list data structure. I will also use results from Mathlib relating to basic arithmetic, divisibility and operations on lists. I will not be using any functions from Mathlib relating to factorisation or prime numbers, they will be formalised explicitly when required. I will likely use the standard induction tactic seen in the Natural Number game.